

Personal Profile

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Professional Profile

An ambitious data scientist with a PhD in statistics-driven environmental microbiology and a strong educational background in mathematics, probability, and statistics. Have a comprehensive understanding and applied expertise in genomic and environmental data analysis, machine learning, bioinformatics, and computational statistics, gained through both academic research and over two years of industry experience as a Data Scientist at Abbott Laboratories. Technically proficient in R, JMP, Linux, SQL, and Bash scripting, with extensive experience in high-resolution time-series data, AI-driven predictive modelling (including Random Forest and LSTM methods), and next-generation sequencing analysis. A calm, reliable, and self-motivated team player with a strong work ethic, capable of working independently to deliver high-quality, insightful work, and motivated to continue developing advanced data-driven solutions in a research-intensive, interdisciplinary environment.

Key Achievements

- Completed a PhD focusing on AI-driven solutions to predicting pathogenic bacteria in the Baltic Sea
- Employed as Data Scientist at Abbott Laboratories for over 2 years
- Completed an MSc. in Computational Genomics – Grade 2.1
- Achieved 1.1 Grade in BSc Thesis and 2.1 overall
- Selected as Mathematics Teacher and Tutor in National University of Ireland Galway (NUIG) for first and second year students

Technical Expertise/Skills Profile

- **Data Analysis** – Extensive experience in real-world data analysis supporting process optimisation, material performance, and yield analysis, as well as PhD-level analysis of high-resolution environmental time-series data. Strong experience in development and evaluation of machine learning models (Random Forest, LSTM (using tensorflow)) for predictive and forecasting applications alongside genomics data analysis and next-generation sequencing workflows, including SAMtools, GATK, BWA, ANNOVAR, RNA-sequencing, and ChIP-seq. Proficient in statistical software packages including Minitab and MATLAB.
- **Microbiology & Genomics** – Comprehensive knowledge of genomics and specifically microbiology, intertwined with data analytics and computer science
- **Mathematics** - Strong competence in mathematics, probability and statistics from university and work experience
- **Programming** - Strong functional and technical skills in numerous languages such as JMP, R, R Shiny, and Bash Scripting, as well as SQL
- **Operating Systems** - Vast experience working in Linux during the duration of my postgraduate studies
- **Problem Solving Skills** – Through numerous failure investigations and participation root cause analysis teams I have developed exceptional problem-solving skills, frequently applied to analyse and interpret different types of data
- **Communication Skills** – Strong verbal and written communication skills, developed through educational and professional employment. Proven ability to communicate with people of all levels through working in an environment with daily meetings, presenting my PhD results in numerous conferences worldwide, and involvement in sport. Highly effective in interpreting, presenting and communicating statistical results to audiences.

Educational and Work Experience

- **Feb. 2022 to Jan 2026** **PhD student, Leibniz Institute for Baltic Sea Research, Rostock, Germany**
- **Thesis:** *Quantification, Prediction and Forecasting of Vibrio vulnificus in the Southern Baltic Sea Using High-Resolution Data*
Submitted (grade pending)
Employed as PhD student in a marine research institute. My role focused on using my background in statistics to implement machine learning methods on environmental biology datasets. In particular I analysed a time-series dataset which I also created using weekly water samples. The goal was to create an early-warning system in the Baltic Sea to detect *Vibrio vulnificus*, which is a bacteria that causes numerous deaths through infection each year in the Baltic Sea. I used R and JMP to achieve this, and implemented numerous machine learning methods, mainly using Random Forest and Long-Short Term Memory models. During my time in Germany my main achievements have been:

- My manuscript “**AI-Driven Forecasting of *Vibrio vulnificus* in the Southern Baltic Sea Using High-Resolution Data**”.

My final first-author manuscript, I did the data gathering initially for this and came up with the idea to use the dataset we gathered for this purpose. Using my background I implemented Random Forest and LSTM models to successfully forecast when this deadly bacteria would be present in the Baltic Sea, and finally wrote this manuscript independently. Submitted to Water Research, which is in the top 2% of water-related journals (impact factor 12.4).

- **Published author of multiple peer-reviewed papers**

Involved in writing and analysis of 4 other papers, one of which I was first author on. Tasks included using datasets to find optimal machine learning methods for specific environmental datasets, assessing the health of the Baltic Sea based on chemical contamination, and creating a quick way to detect flesh-eating bacteria.

- **Creating a high-resolution dataset**

Carried out a hugely comprehensive sampling campaign which involved 15 locations in the Baltic Sea and Warnow Estuary. Sampled 14 locations twice each week by going to each location and taking water, and did numerous biological procedures in laboratory to quantify various biological parameters (https://www.iow.de/otcg_home.html).

- **Nov. 2019 to Jan. 2022 Data Scientist, Abbott Diabetes Care, Donegal, Ireland**

Worked as a data scientist for Abbott Laboratories. Through this role I built on my knowledge of statistics and programming, and learnt a great deal about the real-world application of data analysis. My primary focus was on blood glucose test strips. On a daily basis my job varied from statistical support and programming projects, to ad-hoc failure and root-cause analysis investigations, as well sampling plans, validation and document approvals. In my time in the role of data scientist my key achievements have been:

- **Smart New Analytics Project (SNAP) application**

Creation of a data analytics tools that could be used by a wide range of Abbott personnel. This project involved writing numerous scripts in JMP that would create an interface whereby users could review data and reports. The scripts relating to data preparation would run automatically each day.

- **Data Warehouse Upgrade**

Led a project that upgraded the current SQL database in ADC Donegal. This involved applying for an investment of €30,000 from ADC in order to get assistance from an external company that specialises in software solutions. My own team and an external team then worked in close collaboration to develop the various data warehouses.

- **Test Review Automation**

ADC Donegal do on-site product testing on blood glucose strips, and the data is stored in excel files. Two scripts were created that extracted the data from excel files, and created a report consisting of graphs and tables. Reduced hours for an ADC employee to compile this Test Review from one day to 2 minutes.

- **Lean Sigma Yellow Belt Project**

Predictive model improvement for on-site testing of blood glucose strips. The model is used to predict the results of on-site strip testing using various materials and processes further back in production as predictors

- **Conversion Factor Review**

Analysis of the conversion factor, which is an equation that converts the amount of glucose in blood to a synthetic blood solution that is used in various on-site testing. For this review, data from clinic test sites in England are compared to data from testing within ADC Donegal.

- **Sep. 2018 to Apr. 2019 Teacher, NUI Galway, Ireland**

Teach classes of mathematics and business to first and second year students in tutorials

Prepare students for upcoming exams through teaching exam-based questions

Content included mathematics, applied mathematics, statistics to third level education standard

- **Sep. 2018 to Jul. 2019 Tutor (SUMS: Support for Undergraduate Mathematics and Statistics), NUI Galway, Ireland**

Specialised as a Statistics and Probability Tutor.

Provide a service designed to help students with assignments and tests.

Work on a one-to-one basis to assist students with problems they cannot solve.

Bachelors and Masters Qualifications

- **Sep. 2018 – Aug. 2019 MSc. in Computational Genomics, NUI Galway, Ireland. Grade: 2.1**

Thesis: *Predicting Links in Gene Regulatory Networks from other Networks*

Grade (Irish Grading System): 2.1

Project predicted links in gene regulatory networks of yeast using 5 other genetic networks.

Project was implemented on R-Studio, and the machine learning methods used were:

Logistic Regression, Lasso, Ridge, Elastic Net, SVM, and Decision Forest, with a 10×10- fold cross-validation used for analysing and comparing these methods.

- **Sep. 2014 – May. 2018** **BSc. in Mathematics, NUI Galway, Ireland. Grade: 2.1**
Thesis: *Determinants of Wartime Success*
Grade (Irish Grading System): 1:1
Final year project involved gathering data on conflicts between countries of over 10,000 casualties between 1820 - present. Number of casualties, population, GDP, and number of allies of countries were used as variables for predicting the outcome of the conflicts. Logistic regression on Minitab was used to analyse data.

Additional Skills and Interests
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- Full clean Driving Licence
- Enjoy running and playing sport, particularly Gaelic football.

References

- Educational and work references available on request.